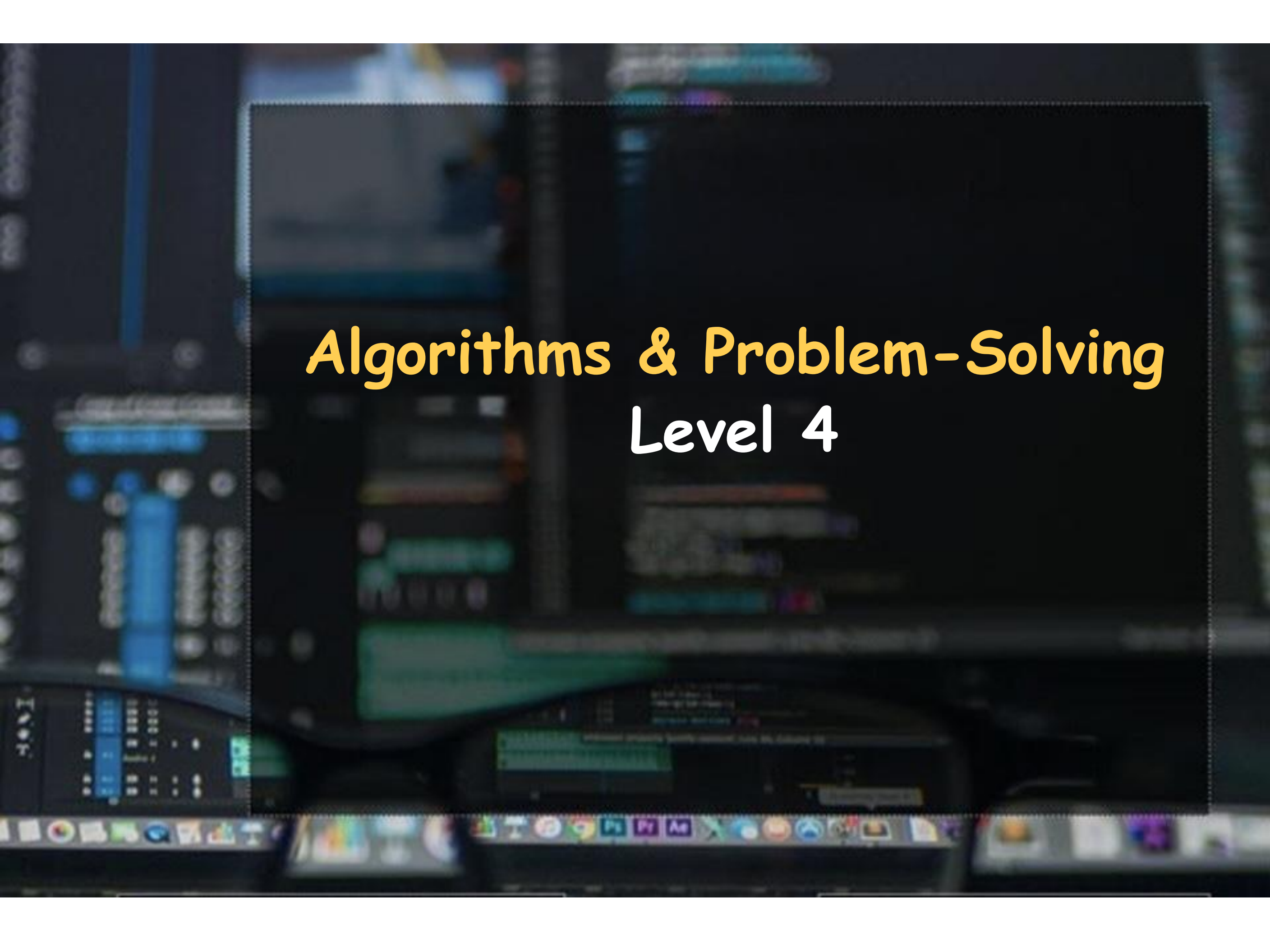
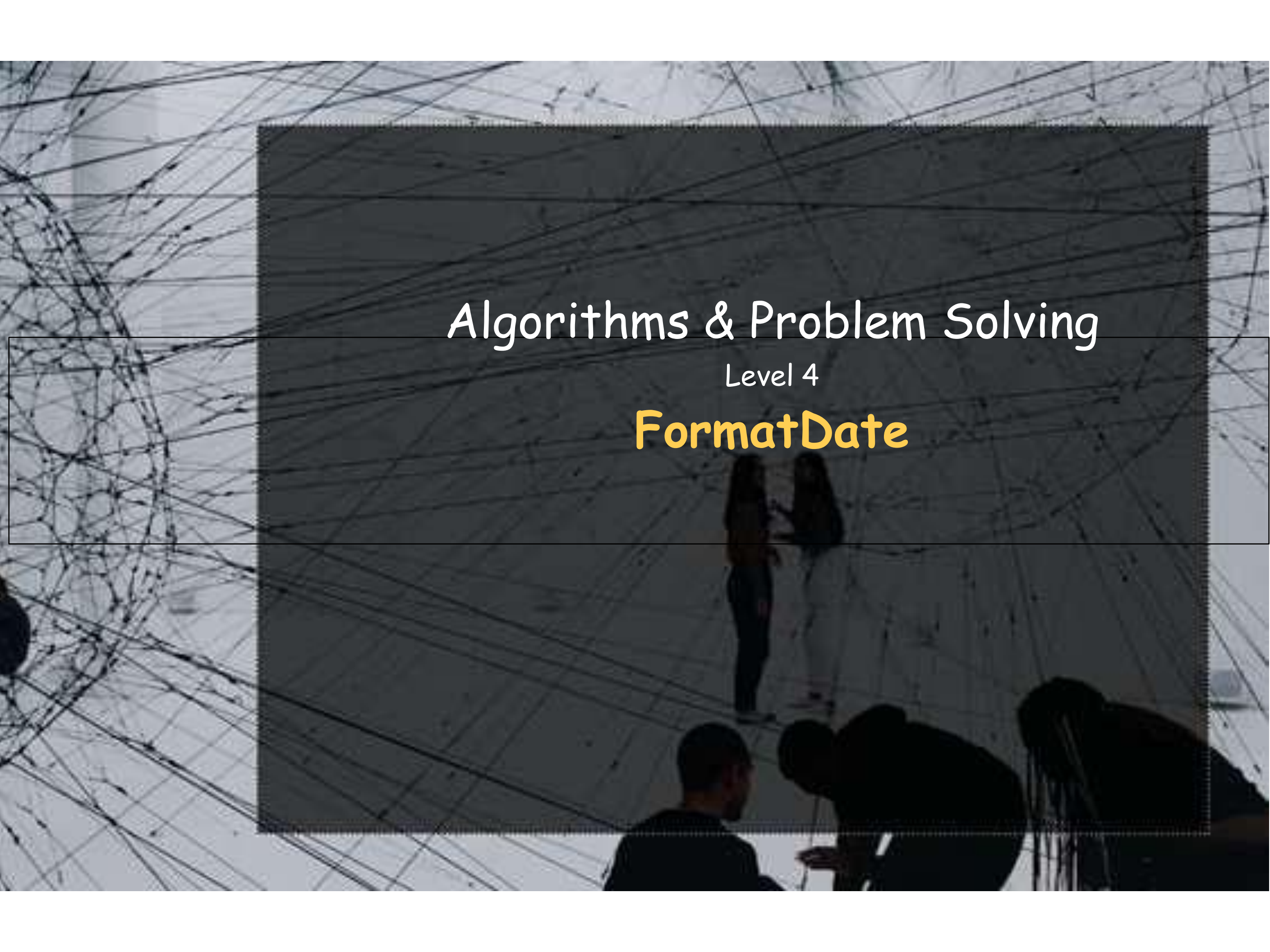




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

FormatDate



Definition

- A program that reads a date from the user in text format, such as 31/12/2022, and then displays it in several different formats depending on the required format, such as:

31/12/2022

2022/31/12

12-31-2022

Day: 31, Month: 12, Year: 2022



Explanation:

- The program's objective is to customize the date display using the `FormatDate` function, which relies on replacing symbols within the text (dd, mm, yyyy) with the day, month, and year values from the `sDate` structure.

Procedures

① struct sDate Definition

- Contains three elements:

 - Day

 - Month

 - Year

② SplitString Function

- Definition: Splits string into parts based on a specific parameter (such as "/")

- 💡 Used to extract the day, month, and year from the input string

③ StringToDate Function

- Definition: Converts the string dd/mm/yyyy to a variable of type sDate

- Depends on the SplitString function and then uses stoi to convert the string to integers

④ DateToString Function

- Definition: Converts the date structure to text in the format "dd/mm/yyyy"

⑤ ReplaceWordInString Function

- Definition: Replaces each occurrence of a word within a string with another word

- 💡 Used here to replace dd, mm, and yyyy with the actual values For Dates

6 The FormateDate Function

- Definition: Creates a date in a user-defined format

- Takes two parameters:

 - sDate date (the date itself)

 - string dateFormat (the desired format, such as "mm-dd-yyyy")

- 📌 Note: If the user does not specify a format, "dd/mm/yyyy" is used as the default format

7 The Main() Function

- Requires the user to enter a date

- Converts it to sDate

- Prints multiple date formats using FormateDate



Example:

Please Enter Date dd/mm/yyyy? 31/12/2022

31/12/2022

2022/31/12

12/31/2022

12-31-2022

31-12-2022

Day:31, Month:12, Year:2022

Summary

- ✓ The program explains the concept of output formatting.
- ✓ It relies on converting between text and structure.
- ✓ It can be used to display dates in different formats according to system or user requirements.

Thank you 😊